



MIST- MEDICAL INSTITUTE FOR SCREENING TEST

BIOCHEMISTRY SUBJECT WISE TEST DATED ON 27 APRIL 2026

QUESTIONS

Q1. VLDL and Chylomicron carry TAG in plasma from liver and small intestine respectively. Which of the following enzyme and its allosteric activator is best known for the selective uptake of TAG in peripheral tissues?

- a) Hormone sensitive lipase & glucagon
- b) Lipoprotein lipase & apo C-II
- c) LCAT & HDL
- d) Cholesterol ester hydrolase & apo C-I

Q2. Hemoglobin acts as a buffer in biological system. Which of the following amino acid protonate and deprotonate at neutral pH and provides buffering action to hemoglobin in blood at pH of 7.4?

- | | |
|--------------|------------|
| a) Histidine | c) Lysine |
| b) Arginine | d) Glycine |

Q3. GM1 is an important ganglioside present in intestine mucosal cell that facilitates the entry of Cholera toxin through gut. This further increases cAMP that disrupts the balance of electrolytes and water, causing diarrhea. Which of the following mechanism best explains the above mechanism?

- a) Phosphorylation of Gs α subunit
- b) ADP Ribosylation of Gs α subunit
- c) Dephosphorylation of Gs β subunit
- d) ADP Ribosylation of Gs γ subunit

Q4. Apo E IV is implicated in the development of :

- a) Alzheimer's disease
- b) Parkinson's disease
- c) Huntington's disease
- d) Primary Familial Hypercholesterolemia

Q5. A 45 year old working female presents to the OPD with complaints of generalized weakness. On admission there was a single episode of vomiting. She is otherwise fine. Further workup reveals Serum total bilirubin 15.6 mg% (conjugated 7.6 % & unconjugated 8.0 %), Serum ALP – 227 IU/L, AST – 3200 U/L, ALT – 2734 U/L, Urine bile salts – Negative but fecal Stercobilinogen is Positive. What is your best probable diagnosis?

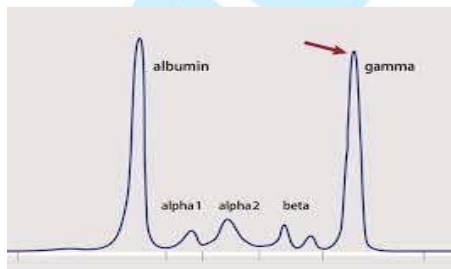
- a) Hemolytic jaundice
- b) Acute viral hepatitis
- c) Obstructive jaundice due to gall stones
- d) Gilbert syndrome

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Q6. Warburg's hypothesis explains that neoplastic cells have a high rate of metabolism due to the fact that they derive its nutrition through angiogenesis that aids in maturation of stromal cells. Which of the following mechanism best explains the above statement?

- a) Increased oxidative phosphorylation by increase in numbers of mitochondria
- b) Neoplastic cells form a fast-food joint
- c) Neoplastic cells undergo spontaneous anaerobic glycolysis
- d) Neoplastic cells possess increased gluconeogenesis

Q7. A 36 year old school teacher presents to the OPD with a history of lower back pain and painful urination. On work up it reveals normocytic normochromic anemia with hypercalcemia. The physician asks for a special urine test in which Bradshaw's test comes positive but Jacobson & Milner test (Heat test) comes negative. To evaluate further serum protein electrophoresis was done that shows the following.



What is the most likely diagnosis?

- a) Monoclonal gammopathy of undetermined significance
- b) Multiple Myeloma
- c) Guillain-Barré syndrome
- d) AIDS

Q8. A freedom fighter is in his 3rd day of starvation refuses even water intake and continues his protest. In this situation survival is mainly attributed to gluconeogenesis in liver. Total no. of high energy phosphates required in him to convert 2 molecules of lactate to 1 molecule of glucose are :

- | | |
|------|------|
| a) 2 | c) 6 |
| b) 4 | d) 8 |

Q9. A 9-year-old boy referred to secondary care with an unusual presentation of a fishy odour to his hands, feet, saliva and urine. On presentation, clinical examination was unremarkable. Laboratory investigations including urine analysis and genetic testing confirmed the diagnosis of trimethylaminuria. The patient was referred to a geneticist and dietician, and consequently treated with dietary modification. This rare defect is caused by deficiency of which enzyme?

- a) Fumarylacetoacetate hydrolase
- b) Methane monooxygenase
- c) Monooxygenase 3 (FMO3)
- d) D Amino acid oxidase



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Q10. All of the following amino acids causes aminoaciduria in Cystinuria except :

- a) Cysteine
- b) Ornithine
- c) Leucine
- d) Arginine

Q11. The only injectable form of vitamin K intravenously is:

- a) Menaquinone
- b) Phylloquinone
- c) Primidone
- d) Menadione

Q12. A 42 year old lady working as a CEO in a MNC presents to Endocrinology OPD with chronic flushing of the face, signs of right sided heart failure and “pellagra.” She was also found to have a non-malignant ovarian teratoma with argentaffin cells in large numbers. After operation all symptoms (except the cardiac) disappeared and the excretion of a metabolite in urine comes back to normal. What is the most likely compound that was earlier excreted in high quantity in Urine?

- a) FIGLU
- b) 3 – OH Anthranilic acid
- c) 5 – OH Indole Acetic Acid
- d) 3,4 Di Hydroxy Phenylalanine

Q13. Compartmentalization is a feature of all of the following pathways except:

- a) Urea cycle
- b) Gluconeogenesis
- c) Heme synthesis
- d) Heme breakdown

Q14. A 46 year old male presents to the OPD with history of progressive muscular dystrophy. Which of the following plasma enzymes will be of diagnostic value?

- a) Pyruvate Dehydrogenase
- b) Lactate Dehydrogenase
- c) Creatine Kinase
- d) Glutamate Dehydrogenase

Q15. Deficiency of LCAT leads to which clinical finding?

- a) High levels of HDL
- b) Corneal opacities



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- c) Hypertriglyceridemia
- d) Tendon xanthomas

Q16. Fate of Pyruvate include all of the following except :

- a) Oxidative decarboxylation to Acetyl CoA being aided by PDH complex
- b) Biotin dependent carboxylation to form oxaloacetate during gluconeogenesis
- c) Deamination to Alanine by deaminase promoting protein anabolism
- d) Anaerobic glycolysis to Lactate predominantly in exercising muscles, cornea, lens, medulla of kidney, testes, WBC & RBC.

Q17. A vasodilating compound is produced by the decarboxylation of which of the following amino acid ?

- a) Arginine
- b) Aspartic acid
- c) Glutamine
- d) Histidine

Q18. Best source of reducing equivalents (NADPH) for the FAS I complex in Liver is :

- a) Kreb's cycle
- b) Glycolysis
- c) HMP Shunt
- d) Malate oxaloacetate shuttle

Q19. ALP level in serum is elevated in all of the following except:

- a) Rickets
- b) Obstructive Jaundice
- c) Myocardial Infarction
- d) Infective hepatitis

Q20. Which of the following enzyme deficiency is most likely to cause the following?



- a) GAL1PUT
- b) Aldose reductase
- c) Sorbitol dehydrogenase



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d) Galactokinase

Q21. In which of the following tissues glucose transport into the cell is insulin sensitive?

- a) Brain
- b) Lens of eye
- c) RBC
- d) Adipose tissue

Q22. All are true about β oxidation of Fatty Acids except :

- a) Activation of FA is a must
- b) Activation requires 2 ATP
- c) Carnitine palmitoyl transferase I is present in outer mitochondrial membrane
- d) Carnitine palmitoyl transferase II is present in mitochondrial matrix

Q23. The process of reconversion of lactate formed in the muscles and RBCs back to glucose in liver cytosol is known as:

- a) Cahill cycle
- b) Cori Cycle
- c) Rapoport Leubering Shunt
- d) Dickens- Horecker cycle

Q24. Increased levels of serum alanine after fasting suggests :

- a) Increased release of alanine from muscles for gluconeogenesis
- b) Reduced amino acid utilization for gluconeogenesis
- c) Break in continuity of plasma membrane due to faulty spectrin and ankyrin resulting in leakage of amino acids
- d) Decreased uptake of alanine by liver

Q25. Compared to the resting state vigorously contracting muscle shows:

- a) An increased conversion of lactate to pyruvate
- b) A decreased NADH/NAD⁺ ratio
- c) An increased concentration of ATP
- d) Decreased level of fructose 2,6 bisphosphate

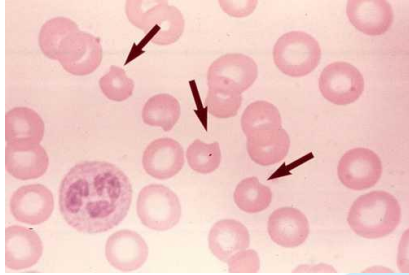
Q26. In preparation for a trip to an area in India where malaria is endemic a young traveller is given antimalarial drug prophylactically. Unfortunately, he immediately develops hemolytic anemia with the following in peripheral



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blood smear. The most likely cause of the hemolysis is a less than normal level of which of the following?



- a) Glucose 6 Phosphate
- b) Oxidized form of $[NADH + H^+]$
- c) Reduced form of glutathione
- d) Ribose 5 phosphate

Q27. A neonate few days after birth suddenly becomes lethargic and starts refusing breast feeding and vomits during forced breast feeding. However, accepts the glucose-water. Mother states that the baby had developed diarrhoea on the 3rd day. Clinical examination reveals that the infant has developed pot belly due to hepatomegaly. Urine R/E reveals presence of Galactose in Urine. Which diagnostic modality was used?

- a) Shilling's test
- b) Rothera's test
- c) Benedict's test
- d) Oxidase-peroxidase test

Q28. Which of the following is a diagnostic test may be used in Maple Syrup Urine disease?

- a) Benedict's test
- b) Dinitrophenylhydrazine test
- c) Ferric chloride test
- d) Cyanide nitroprusside test

Q29. "All enzymes are not proteins" – The supportive statement of this is :

- a) All enzymes do not follow the Michaelis Menten hypothesis
- b) RNAs act as ribozymes
- c) Antibodies take part in the catalysis of many reactions
- d) Metals are involved in attachment to enzymes and catalysts

Q30. All of the following are true regarding collagen except :

- a) Triple helical structure is a right handed superhelix
- b) Glycine residue at every 3rd position of individual α chains
- c) Vitamin C is necessary during Post translational modification
- d) Lysyl hydroxylase is needed for effective collagen cross linking



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Q31. Protein 3D structures are best studied by :

- a) SDS-PAGE
- b) Tandem Mass Spectrometry
- c) X-Ray diffraction Crystallography
- d) CD Spectroscopy

Q32. All of the following enzymes contribute to Substrate level phosphorylation except :

- a) Phosphoglycerate kinase
- b) Phosphofructokinase
- c) Succinate thiokinase
- d) Pyruvate kinase

Q33. A 46 year old female presents returns from China. At the airport she experiences shortness of breath and mild cough. Eventually she was stopped at covid screening at airport and was sent into quarantine. RT PCR comes positive. On workup total LDH was also found to be raised in plasma. Further in serum protein electrophoresis which of the following isoenzyme of LDH is most likely to be elevated in this patient?

- a) LDH1
- b) LDH2
- c) LDH3
- d) LDH 4

Q34. Detoxification of ammonia in brain is important as it can even lead to coma. Ammonia is transported from brain as:

- a) Alanine
- b) Aspartate
- c) Glutamate
- d) Glutamine

Q35. In Reye's syndrome there is triad of hyperammonemia, metabolic acidosis & convulsion in a child. Which of the following mechanism best explains the pathogenesis behind this disorder?

- a) Mitochondrial damage leading to escape of Ornithine Transcarbamoylase into cytosol
- b) Decrease in UMP synthase in Pyrimidine synthesis
- c) Deficiency of Ornithine Transcarbamoylase in Urea cycle
- d) Deficiency of Carbamoyl phosphate synthetase-1 in Urea cycle

Q36. Which enzyme is active during low insulin glucagon ratio?

- a) Hexokinase
- b) Glucokinase
- c) Glucose 6 Phosphatase
- d) Phosphofructokinase



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- Q37. Which of the following is a branching enzyme ?
- Glycogen phosphorylase
 - Amylo 4,6 glucosidase
 - Amylo 1,4-1,6 transglycosylase
 - Glucosyl 4,6 transferase
- Q38. Which of the following amino acid enters TCA cycle for gluconeogenesis but also ketogenic in nature ?
- Methionine
 - Arginine
 - Histidine
 - Tryptophan
- Q39. The biosynthesis of the enzyme pyruvate carboxylase is repressed by :
- Insulin
 - Cortisol
 - Glucagon
 - Thyroxine
- Q40. In a well fed state the activity of the CAT-1 in outer mitochondrial membrane is regulated by :
- Blood Glucose
 - Acetyl CoA obtained from β oxidation of LCFA itself
 - Pyruvate obtained from PDH multi enzyme complex
 - Malonyl CoA obtained from Acetyl CoA carboxylase
- Q41. All of the following has the potential to cause Homocystinuria except :
- Methionine synthase enzyme hypoactivity due to less Vit B12 intake in diet
 - Cystathionine β synthase deficiency & lack of Vit B6 intake in diet
 - Point mutation in Methylene Tetra Hydro Folate Reductase
 - Deficiency of Cystathionine γ lyase causing decrease in Transsulfuration
- Q42. Glucose can be synthesized from all except :
- Alanine
 - Acetyl CoA
 - Lactate
 - Glycerol



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Q43. A person had his last night meal at 8 PM as next morning he was supposed to give blood sample for fasting plasma glucose. Next morning at 9 AM his FPG was recorded to be 186 mg/dL. What's the source of this said value of glucose?

- a) Dietary glucose
- b) Hepatic gluconeogenesis
- c) Hepatic glycogenolysis
- d) Muscle glycogenolysis

Q44. All of the following glycolytic enzymes are under the control of insulin except :

- a) Phosphofructokinase – I
- b) Hexokinase
- c) Phosphoglycerate kinase
- d) Pyruvate kinase

Q45. Aldolase can be considered as a lyase for which of the following substrates ?

- a) Glucose 6 phosphate
- b) Fructose 6 Phosphate
- c) Fructose 1 Phosphate
- d) Glucose 1 phosphate

Q46. Enzyme common to both Glycogenesis and Glycogenolysis :

- a) Glucokinase
- b) Glucosyl 4,6 transferase
- c) α 1,6 glucosidase
- d) Phosphoglucomutase

Q47. Glucagon receptors are absent in :

- a) Brain
- b) Cornea
- c) Adrenal gland
- d) Kidney

Q48. A 26 year old male presents with intense muscle cramps, fatigue, and exercise intolerance following an intense exercise in the gym. high-carbohydrate meals often worsen the symptoms and severe exercise presents with patient passing burgundy coloured urine. 'Second wind phenomenon' is not reported though. What is the most likely underlying disorder?

- a) Type 0 glycogen storage disease
- b) Type V glycogen storage disease
- c) Type VI glycogen storage disease
- d) Type VII glycogen storage disease



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Q49. Krabbe's disease is due to the deficiency of :

- a) Sphingomyelinase
- b) β glucosidase
- c) β galactosidase
- d) α galactosidase A

Q50. The most common lysosomal storage disorder that is mostly treatable by enzyme replacement therapy is :

- a) Wolman's disease
- b) Gaucher's disease
- c) Norum's disease
- d) Tay Sach's disease

Q51. In Type I hyperlipoproteinemia there is a defect in :

- a) Apo A-I
- b) Apo A-II
- c) Apo C-II
- d) Apo B-100

Q52. You have consumed a high fat rich meal in the night just before your examination. Acetyl CoA which is the connecting link of entire metabolic pathways of proximate principle during overnight sleep it is most likely to form ?

- a) Palmitoyl CoA
- b) Succinyl CoA
- c) Malonyl CoA
- d) Acetoacetyl CoA

Q53. False regarding medium chain fatty acid :

- a) Predominantly found in ghee, butter, milk
- b) Doesn't require pancreatic lipase
- c) Doesn't get deposited in adipose tissue
- d) Requires carnitine shuttle for its oxidation

Q54. Which of the following lipoproteins is not included in the Frederickson's classification of hyperlipoproteinemia?

- a) VLDL
- b) IDL
- c) LDL
- d) HDL



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Q55. Match the metabolic urine disorders with their classic urine abnormalities :

LIST 1	List 2
1. PKU	A. Sulfur Odor
2. Indicanuria	B. Sweaty feet Odor
3. Cystinuria	C. Orange sand in diaper
4. Alkaptonuria	D. Mousy Odor
5. Lesch Nyhan syndrome	E. Black colour
6. Isovaleric aciduria	F. Blue colour

- a) 1-D, 2-F, 3-A, 4-E, 5-C, 6-B
- b) 1-D, 2-A, 3-F, 4-E, 5-C, 6-B
- c) 1-D, 2-F, 3-A, 4-C, 5-E, 6-B
- d) 1-B, 2-F, 3-A, 4-E, 5-C, 6-B

Q56. Amino acid linking urea cycle and kreb's cycle is :

- a) Aspartate
- b) Asparagine
- c) Glutamate
- d) Fumarate

Q57. Amino acid needed for glycation of hemoglobin is :

- a) Glycine
- b) Glutamate
- c) Valine
- d) Leucine

Q58. Biopterin supplementation can be considered in treating :

- a) Hyperglycinemia
- b) Alkaptonuria
- c) Phenylketonuria
- d) Hypopigmentation

Q59. A 46 year old chronic alcoholic male was rushed to ER with acute onset abdominal pain, episodic vomiting and lethargy. On examination his electrolytes were normal but shows TAG : 2928 mg/dL & GGT 2008 IU/L with De Ritis ratio < 2 , Serum Amylase : 112 IU/L, Lipase : 918 IU/L. CECT was not available. Which of the following may be considered as an special test for confirmation of diagnosis?

- a) LDH1
- b) LDH2
- c) LDH3
- d) LDH4



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Q60. Biochemical process that releases urea is:

- a) Carboxylation
- b) Decarboxylation
- c) Thiolytic
- d) Hydrolysis

Q61. Selenocysteine is found in all of the following except:

- a) Deiodinase
- b) Glutathione reductase
- c) Thioredoxin reductase
- d) Glutathione peroxidase

Q62. Serum ferroxidase level decreases in:

- a) Gaucher's disease
- b) Cirrhosis of liver
- c) Acute pancreatitis
- d) Wilson's disease

Q63. A chronic alcoholic presents to clinic with generalized weakness and lethargy. He was also telling stories that was not entirely true as per the accompanying person. On examination intern on duty also elicits global confusion, nystagmus, ataxia. What's the most specific treatment for this?

- a) Injection of Vitamin B2
- b) Injection of Vitamin B9
- c) Injection of Vitamin B1
- d) Injection of Vitamin B12

Q64. A 10 yr old child with aggressive behavior and poor concentration to studies is brought with present complaints of joint pain especially in lower limb and reduced urinary output. Mother however gives h/o self-mutilative behavior by the child stating that he tends to mutilate his fingers by biting. Which of the following enzyme is likely to be deficient in this child?

- a) HGPRTase
- b) Adenosine deaminase
- c) APRTase
- d) Acid maltase

Q65. Best technique to detect post translational modification of enzymes is:

- a) Southern blot
- b) Northern blot



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- c) Western blot
- d) Eastern blot

Q66. A genetic disorder renders fructose 1,6 biphosphatase in liver less sensitive to regulation by fructose2,6 biphosphate. All of the following metabolic changes are seen except :

- a) Less pyruvate is formed
- b) Less ATP is generated
- c) Level of Fructose1,6 biphosphate will be lower than normal
- d) Level of Fructose1,6 biphosphate will be higher than normal

Q67. Which of the following vitamin deficiency is evident with obstructive jaundice ?

- a) Vitamin B3
- b) Vitamin B12
- c) Vitamin C
- d) Vitamin K

Q68. Lp(a) excess is best treated by :

- a) Thiamine
- b) Niacin
- c) Pyridoxine
- d) Methylcobalamin

Q69. Conversion of nor adrenaline to adrenaline is done by :

- a) Addition of a methyl group
- b) Removal of CO₂
- c) Addition of NH₂ group
- d) Removal of a methyl group

Q70. PCR technique uses :

- a) Mn²⁺
- b) Mg²⁺
- c) Ca²⁺
- d) Fe²⁺

Q71. Severe thiamine deficiency is associated with :

- a) Decreased RBC transketolase activity
- b) Increased clotting time
- c) Decreased RBC transaminase activity
- d) Increased xanthurenic acid excretion



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Q72. A 17-year old boy presents to medicine OPD with staining of teeth and occasional blister formation when exposed to sunlight for a long time. The medicine HOD refers the case to Dermatology which further gets referred to Biochemistry. Laboratory reports reveal raised Coproporphyrin-I levels and explains the increased risk of photosensitivity and diagnose the case to be one rare form of porphyria. Most likely enzyme deficient is :



- a) Uroporphyrinogen I Synthase
- b) Uroporphyrinogen III Synthase
- c) Uroporphyrinogen Decarboxylase
- d) Coproporphyrinogen Oxidase

Q73. A child presents with recurrent infections. Laboratory tests reveal low levels of both B and T lymphocytes. Genetic testing shows a mutation in the adenosine deaminase (ADA) gene. What is the most likely diagnosis?

- a) X-linked agammaglobulinemia
- b) Common variable immunodeficiency (CVID)
- c) Severe combined immunodeficiency (SCID)
- d) DiGeorge syndrome

Q74. Biotin act as a coenzyme for all except :

- a) Pyruvate to Oxaloacetate
- b) Acetyl CoA to Malonyl CoA
- c) Propionyl CoA to Methyl Malonyl CoA
- d) Glutamate to Gamma Carboxy Glutamate

Q75. Transport of neutral amino acids through membranes of the intestine, kidney & Liver is mediated by :

- a) Kreb's cycle
- b) Cori's cycle
- c) Meister cycle
- d) Cahill Cycle

Q76. A 53-year-old woman comes to the physician because of a 6-month history of diffuse weakness and tingling of her arms and legs. CNS examination shows loss of power in extensor and flexor muscles of the lower extremities.



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Sensation to vibration and position is decreased in all extremities with positive Romberg sign. This patient most likely has a deficiency of which of the following vitamin?

- a) Vitamin B1
- b) Vitamin B2
- c) Vitamin B6
- d) Vitamin B12

Q77. A 29 year old man on a trekking expedition to western ghats gets stranded in deep forest without any food supplies however, luckily was rescued after a week. On return during a health check at a camp his blood ketone bodies were moderately high. Which of the following cannot use these ketone bodies for its energy supply?

- a) Brain
- b) RBC
- c) Skeletal muscles
- d) Adipocytes

Q78. Which of the following food item is highest thermogenic?

- a) Carbohydrate
- b) Protein
- c) Fat
- d) A mixed diet of all 3 in equal amount

Q79. A 9 year old male child was brought to the pediatric OPD with coarse facial hair, protuberant abdomen, frontal head enlargement and thickening of cardiac valves, hepatosplenomegaly and hearing impairment. What is the most likely diagnosis?

- a) Hurler syndrome
- b) Hunter syndrome
- c) Niemann pick disease
- d) Tay sach's disease

Q80. Positive ferric chloride test during screening of phenylketonuria is due to urinary excess of :

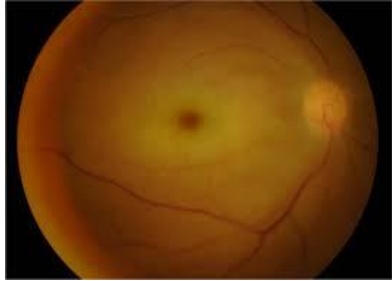
- a) Phenyl pyruvate
- b) Phenyl acetate
- c) Phenyl lactate
- d) Phenylalanine

Q81. Following is not seen in :



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- a) Niemann Pick's disease
- b) Gaucher's disease
- c) Tay Sach's disease
- d) Krabbe's disease

Q82. Mitochondria is involved in all of the following except:

- a) ATP production
- b) Apoptosis
- c) Kreb's cycle
- d) Cholesterol synthesis

Q83. Patient presents with drying of skin and appears as shown below. Family members also complains of the patient passing extremely foul smelling stool. Recently he started forgetting past events as well.



Which of the following dietary advice that you will suggest for this patient ?

- a) Avoid consumption of meat products
- b) Avoid eating polished rice
- c) Avoid high gluten diet
- d) Avoid eating maize

Q84. A 25 year male presents with hypercholesterolemia and increased risk of heart disease. His elder brother also had similar findings and succumbed to this. Workup reveals surprisingly a normal triglyceride and VLDL level in plasma and finally the disorder was identified to be due to a defect of congenital LDL receptor deficiency. The most likely defect will be?



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- a) Primary familial hypercholesterolemia
- b) Abetalipoproteinemia
- c) Familial combined Hyperlipidemia
- d) Tangier's disease

Q85. A 40 year old male presents to the Medicine OPD with recent complaints of poor concentration to his work, frequent forgetfulness of the works assigned to him and slow bodily movements since he had attended a friend's marriage almost 3 weeks back. Workup revealed the case to be a form of prion disease primarily noted in human due to consumption of infected meat. What is the most likely condition?

- a) Alzheimer's disease
- b) Parkinson's disease
- c) Creutzfeldt Jakob disease
- d) Fatal Familial insomnia

Q86. All of the following are true regarding Ehler's Danlos syndrome except:

- a) Patient may present with sensorineural hearing loss
- b) There is hyper stretchability of skin
- c) Patient can develop ecchymoses with minor trauma
- d) In severe cases there might be aortic aneurysms noted

Q87. Iron absorption through our GIT is stimulated by keeping the iron in its ferrous form. One of the following is playing a major role in decreasing the absorption of iron by opposing this mechanism:

- a) Vitamin A
- b) Vitamin C
- c) Oxalates
- d) Gluten

Q88. Which of the following enzyme activity is decreased in fasting?

- a) Hormone sensitive lipase
- b) Glycogen phosphorylase
- c) Acetyl CoA carboxylase
- d) Pyruvate carboxylase

Q89. During exercise fastest way to synthesize ATP is:

- a) Muscle glycogenolysis
- b) Muscle glycolysis
- c) Muscle creatine phosphate
- d) Muscle mitochondrial Krebs's cycle



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Q90. All of the following should be avoided in a lactose intolerant patient except :

- a) Yoghurt
- b) Skimmed milk
- c) Condensed milk
- d) Ice cream

Q91. A 4 months old child was presented to the pediatric OPD with history of irritability, developmental delay and urine burnt sugar smell. Excess of all of the following amino acids can be tested in plasma except :

- a) Leucine
- b) Valine
- c) Phenylalanine
- d) Isoleucine

Q92. An infant present to OPD with following abnormalities as seen in the picture depicted below.



What's the most likely deficient enzyme that would initiate the pathogenesis of ophthalmologic finding in this infant?

- a) Galactokinase
- b) Galactose 1 Phosphate Uridyl Transferase
- c) Aldose Reductase
- d) UDP Galactose 4 epimerase

Q93. Regarding HMP shunt false statement is :

- a) Occurs in cytoplasm and occurs as an alternate of glycolysis
- b) No ATP is produced and pathway is diverted towards formation of reduced glutathione
- c) Active in RBC, gonads, liver
- d) Oxidative phase generates $\text{NADH} + \text{H}^+$ and non-oxidative phase generates pentose phosphates

Q94. A person on a fat free carbohydrate rich diet continues to grow obese and eventually develops stage 1 fatty liver. Which of the following lipoprotein is most specific that is increased in his lipid profile?

- a) LDL
- b) HDL
- c) VLDL
- d) Triglycerides



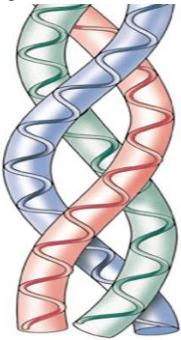
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Q95. Aconitase of Kreb's cycle is blocked by fluoroacetate non competitively. The changes in the kinetic parameters in the presence of fluoroacetate will be reflected as :

- a) Decreased V_{max}
- b) Decreased K_m
- c) No change in V_{max}
- d) Decreased V_{max} and K_m both

Q96. Which vitamin is needed for the post translational modification of the following protein?



- | | |
|---------------|---------------|
| a) Vitamin A | c) Vitamin B2 |
| b) Vitamin B1 | d) Vitamin C |

Q97. The reasons for ketosis in von Gierke's disease are all except:

- a) Acetyl CoA concentration is low
- b) Oxaloacetate is necessary for gluconeogenesis
- c) Blood glucose less than 40 mg/dL
- d) Fatty acid mobilization is high

Q98. All of the following disorders are true regarding use of Anti Cardiolipin antibody test in detecting the pathogenesis except:

- | | |
|-----------------|-------------|
| a) Tuberculosis | c) Cholera |
| b) Malaria | d) Syphilis |

Q99. During the management of Classical Phenylketonuria the all of the following mode can be tried except :

- a) Replacement of the defective enzyme
- b) Replacement of the deficient product
- c) Limiting the substrate for deficient enzyme
- d) Supplementation of bipterin

Q100. The following tube is used in phlebotomy for collection of plasma parameter aiding in diagnosis of diabetes mellitus. Which of the enzyme is being blocked in estimation of the required parameter?



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- a) Glucokinase
b) Pyruvate kinase
c) Enolase
d) Aldolase

ANSWERS

1	b
2	a
3	b
4	a
5	b
6	c
7	b
8	c
9	c
10	c
11	d
12	c
13	d
14	c
15	b
16	c
17	d
18	c
19	c
20	c

21	d
22	d
23	b
24	a
25	b
26	c
27	c
28	b
29	b
30	d
31	c
32	b
33	c
34	d
35	a
36	c
37	d
38	d
39	a
40	d

41	d
42	b
43	c
44	c
45	c
46	d
47	b
48	d
49	c
50	b
51	c
52	b
53	d
54	d
55	a
56	a
57	c
58	c
59	d
60	d

61	b
62	d
63	c
64	a
65	d
66	d
67	d
68	b
69	a
70	b
71	a
72	b
73	c
74	d
75	c
76	d
77	b
78	b
79	b
80	a

81	d
82	d
83	d
84	a
85	c
86	a
87	c
88	c
89	c
90	a
91	c
92	b
93	d
94	c
95	a
96	d
97	a
98	c
99	a
100	c